TEXORA · COURSE NOTES · LECTURE SHEET

Elementary Idea about the Objects of Each Machine Used in Textile Processing

Every machine in a textile mill exists for a specific purpose. This sheet gives the object — the job each machine is there to do — for the principal machines of spinning, weaving, knitting, wet processing and garments manufacturing.

TE-1109 · FUNDAMENTALS OF TEXTILE ENGINEERING · TOPIC 3 · LEVEL 1 TERM 1

1. What “object of a machine” means

A textile mill contains hundreds of machines, and no two of them do exactly the same work. The **object** of a machine is the specific change it is designed to bring about in the material that passes through it: opening the fibre, removing the trash, joining the pieces, applying the colour, and so on. If the object of each machine is understood, the whole mill becomes easy to read — the material flows from machine to machine, and at every step exactly one planned change happens to it.

Two cautions belong here. First, a machine usually has one *main* object and several supporting actions; the card, for example, mainly individualises and cleans the fibres, but it also blends them and forms the sliver. Second, machines are arranged in a fixed order because each one prepares the material for the next: a machine’s output form (lap, sliver, roving, beam, grey fabric) is exactly the input form the following machine is built to accept.

2. Machines of spinning

MACHINE	OBJECT — WHAT IT IS THERE TO DO
Bale opener / blowroom line	To open the compressed fibre from bales into small tufts, remove a large share of the trash, dust and heavy impurities, blend fibres from different bales evenly, and deliver a uniform supply of opened fibre (as a lap or through an air duct) to the carding machines.

Carding machine	To individualise the fibres — separating tufts into single fibres — remove most of the remaining trash, dust and short fibres (neps are opened or removed), begin to parallelise the fibres, and condense them into a continuous, soft rope of fibres called sliver, delivered coiled in a can.
Comber (optional stage)	To remove the short fibres (as noil) from the sliver and retain only the long, parallel fibres, producing the smoothest, strongest and finest yarns. Used for high-quality yarns only.
Draw frame	To improve the uniformity of the sliver by doubling — feeding several slivers together so their thick and thin places average out — and drafting them, which straightens and parallels the fibres, and to deliver one uniform sliver.
Roving frame (simplex)	To draft the sliver down to a much finer strand (roving), insert a small amount of twist so the weak strand can be handled and unwound, and wind the roving onto bobbins of a size the ring frame can use.
Ring frame	To complete the spinning: draft the roving to the final yarn fineness (count), insert the full amount of twist that gives the yarn its strength, and wind the finished yarn onto small bobbins (cops).
Rotor spinning machine (open-end)	To spin yarn directly from sliver in one step — drafting the fibres in an opening roller and reassembling and twisting them inside a fast-spinning rotor — producing yarn for medium and coarse counts without roving or ring frames.
Winding machine (autoconer)	To rewind yarn from the small spinning cops onto large cones or cheeses suitable for weaving, knitting or sale, clearing out faults (thick places, thin places, weak spots) on the way and joining the ends by splicing or knotting.

3. Machines of weaving preparation and weaving

MACHINE	OBJECT — WHAT IT IS THERE TO DO
Winding machine	To transfer warp yarn from spinning packages to large, uniform winding packages with the faults removed, so that warping runs without breaks.
Warping machine	To wind a fixed number of warp ends (a sheet of parallel yarns) from many packages onto a warper’s beam, every end at the same tension and length, in the exact count and order the fabric requires.
Sizing machine	To apply a protective coating of size (traditionally starch paste) to the warp sheet, dry it, and wind the sized warp onto the weaver’s beam — so that the yarn can survive the friction and repeated tension of weaving without breaking.

Loom	To interlace warp and weft yarns at right angles to form woven fabric, by repeating the motions of shedding, picking and beat-up, while letting off warp and taking up cloth at controlled rates.

Shuttleless looms differ only in how the weft is carried across: the **projectile** loom grips and shoots a small projectile carrying the weft; the **rapier** loom carries the weft across on a tape or rod and hands it over in the middle; the **air-jet** loom blows the weft across on a jet of compressed air; the **water-jet** loom does the same with water and is used for filament yarns that water cannot harm. The object is identical in every case — insert one pick, correctly, thousands of times an hour.

4. Machines of knitting

MACHINE	OBJECT — WHAT IT IS THERE TO DO
Circular knitting machine	To form weft-knitted fabric as a continuous tube, using needles arranged in a rotating cylinder fed by many yarn feeders — the main machine for T-shirt, jersey, fleece and hosiery fabrics.
Flat-bed knitting machine	To knit flat fabric panels (and shaped garment pieces) with needles arranged in a flat bed — the main machine for sweaters and fully-fashioned knitwear.
Warp knitting machine (tricot / raschel)	To form warp-knitted fabric from a sheet of warp yarns, each needle supplied by its own yarn — producing stable, run-resistant fabrics such as lace, net, and sportswear fabrics.

5. Machines of wet processing

Wet processing machines work on fabric in rope or open-width form, applying water, chemicals, dyes and heat. The main machines, in the order the fabric usually meets them:

MACHINE	OBJECT — WHAT IT IS THERE TO DO
Singeing machine	To burn off the protruding fibre ends (fuzz) from the fabric surface by passing it rapidly over gas flames or heated plates, so the fabric becomes smooth and prints and dyes evenly.
Desizing, scouring and bleaching range	To prepare the fabric: remove the size applied to the warp (desizing), wash out natural oils, waxes and dirt (scouring), and destroy the natural colouring matter (bleaching), leaving the fabric clean, absorbent and white, ready to take dye uniformly.
Mercerising machine	To treat cotton fabric with concentrated caustic soda under tension, which swells the fibres and gives the fabric greater lustre, strength and dye uptake.

Jigger	To dye fabric in open width by passing it back and forth through a stationary dye bath from one roller to another — suited to woven fabrics that must not be creased.
Winch (becke)	To dye fabric in rope form, circulated gently through the dye bath by a rotating winch — suited to delicate and knitted fabrics.
Jet dyeing machine	To dye fabric in rope form propelled by a jet of dye liquor itself, giving rapid, level dyeing with low liquor usage — the standard machine for polyester and knitwear.
Padding mangle	To impregnate fabric evenly with a solution (dye or finishing chemical) by dipping it in a trough and squeezing it between rollers to a controlled pick-up.
Printing machine	To apply colour in a pattern to defined areas of the fabric — by engraved rollers, flat or rotary screens, or digital inkjet — instead of dyeing the whole fabric one colour.
Stenter	To dry and heat-set fabric held out to a fixed width by clips or pins on a moving chain, controlling width, GSM and shrinkage, and to cure applied finishes.
Calender	To pass fabric between heavy polished rollers under pressure and heat, flattening the yarns to give a smooth, lustrous surface finish.
Compacting / sanforising machine	To pre-shrink knitted or woven fabric mechanically, so that garments made from it do not shrink further in washing.

6. Machines of garments manufacturing

MACHINE	OBJECT — WHAT IT IS THERE TO DO
Spreading machine	To lay fabric from the roll into flat, even layers (a lay) of the marker length, without tension or distortion, ready for cutting.
Cutting machines (straight knife, band knife, CNC cutter)	To cut through the whole lay at once, following the marker lines, producing accurate garment pieces in bundles.
Lockstitch sewing machine	To join fabric pieces with the lockstitch (formed by a needle thread and a bobbin thread interlocking inside the fabric) — the basic machine for main seams of shirts, trousers and most garments.
Overlock machine	To trim the fabric edge and enclose it with a wrapping stitch in one operation, preventing fraying and joining knitwear seams with an elastic stitch.
Flatlock machine	To join knitwear pieces with a flat, smooth seam (butt joint covered by threads) that lies comfortably against the skin — used in T-shirts, underwear and

	sportswear.
Buttonhole and button-sewing machines	To make and finish buttonholes of exact size, and to attach buttons at measured positions, automatically and uniformly.
Bartack machine	To reinforce stress points — pocket corners, belt loops, placket ends — with a dense block of stitches.
Pressing equipment (steam iron, form finisher)	To remove creases, set the garment's shape with steam, heat and pressure, and give the finished product its final appearance before inspection and packing.

KEY POINT

Read any machine through its object: ask what the material looks like before the machine and what it must look like after it. The difference between the two states is the object of the machine — and nearly every machine fault shows up as a failure to make exactly that change.

7. Reading the mill as one system: the material forms

The surest way to test whether the objects are understood is to follow the material and name its form at each machine boundary — every machine of §2–§6 exists to perform exactly one of these conversions:

CONVERSION	PERFORMED BY
Bale fibre → opened, cleaned fibre	Blowroom line
Opened fibre → sliver	Card
Sliver → roving	Draw frame, then roving frame
Roving → yarn	Ring frame / rotor frame
Yarn → warp beam + weft supply	Winding, warping, sizing
Warp + weft → grey fabric	Loom
Yarn → grey fabric (directly)	Knitting machine
Grey fabric → dyed, finished fabric	Wet processing range
Finished fabric → garment pieces	Spreading and cutting machines

Pieces → finished garment	Sewing machines, then pressing

If any conversion is unclear, return to that machine's row in the tables above: its "object" column is precisely the conversion written as a sentence.

8. Glossary — key terms of this topic

TERM	MEANING
Object of a machine	The specific change a machine is designed to make in the material passing through it.
Noil	The short fibres removed by the comber; sold as a by-product for coarse yarns and other uses.
Cop	The small bobbin of yarn wound by the ring frame.
Weaver's beam	The beam carrying the sized warp, mounted on the loom for weaving.
Liquor ratio	The ratio of the weight of dye or processing liquor to the weight of fabric treated in it.
Pick-up	The percentage of solution a fabric takes up in padding, expressed on the weight of the fabric.
Lay	The stack of fabric layers spread for cutting.
Lockstitch	The commonest sewing stitch, formed by interlocking a needle thread with a bobbin thread inside the material.

9. Key points and summary

- The object of a machine is the one planned change it makes in the material; mills are organised so that these changes happen in a fixed, logical order.
- Spinning machines progressively open, clean, parallelise, draft and twist: blowroom → card → (comber) → draw frame → roving frame → ring frame → winding.
- Weaving preparation protects and organises the warp (winding, warping, sizing); the loom's object is one accurate pick at a time, however the weft is inserted.
- Knitting machines form loops: circular machines make fabric tubes, flat beds make panels, and warp knitting machines make stable fabrics from a yarn sheet.
- Wet processing machines prepare (singeing, desizing, scouring, bleaching), colour (jigger, winch, jet, printing) and finish (stenter, calender, compacting) the fabric.

- Garment machines spread, cut, sew (lockstitch, overlock, flatlock and special-purpose machines) and press the pieces into a finished product.

10. Practice

Answer in your own words; brief model answers follow.

1. State the object of the card in one sentence, and name the material form entering and leaving it.
2. A fabric shows colour only in printed motifs, white elsewhere. Which wet-processing machine produced this, and what is its object?
3. Why does the weaving route need a sizing machine but the knitting route does not? Answer through the objects of the machines involved.
4. A garment factory buys an overlock machine and a flatlock machine in the same week. State the different object of each.
5. Place in process order: stenter, ring frame, loom, cutting machine, draw frame — and give the material form between each pair.
6. A mill wants softer, pre-shrunk knit fabric that will not shrink in the customer's wash. Which finishing machine's object matches this need?

Answer key

1. Object: to individualise, clean and parallel the fibres and form them into sliver. In: opened fibre mat/lap; out: carded sliver.
2. The printing machine; its object is to apply colour to selected areas of the fabric only, instead of dyeing the whole piece.
3. Sizing's object is to coat warp yarn with a protective film so it survives loom abrasion; knitting forms fabric by looping at low tension, so no such protection is needed.
4. Overlock: to trim, wrap and secure fabric edges against fraying while joining. Flatlock: to butt-join knit pieces with a flat, comfortable seam.
5. Draw frame (sliver) → ring frame (yarn) → loom (grey woven fabric) → stenter (finished, width-set fabric) → cutting machine (garment pieces).
6. The compacting (sanforising) machine — its object is to pre-shrink the fabric mechanically so later washing causes minimal further shrinkage.